

Python 2: Visualization and Analysis - Assignments

Python 2 – Part 2 (Data Analysis with Python)

Exercise 1 – Linear Regression

In this exercise you will practice performing linear regression using the Fama and French Three Factor Model, an asset pricing model that expands on the Capital Asset Pricing Model (CAPM) by adding size risk and value risk factors to the market risk factor in the CAPM. This model considers that value and small-cap stocks outperform markets on a regular basis.

The general equation for the Fama-French model is:

$$r_p - r_f = \alpha + \beta_1 MRP + \beta_2 SMB + \beta_3 HML$$

Where: $r_p - r_f$ is the expected excess return of the portfolio or stock over the risk free rate; MRP is the market risk premium, SMB is the size premium, HML is the value premium, and $\beta_{1,2,3}$ refer to the factor coefficients.

1. Load the Apple stock data (aapl.csv) and the Fama-French 3 Factor Daily data (ff3_daily.csv found in the ExData folder).
2. Calculate the daily simple returns for Apple using the adjusted close prices. Remember to scale the returns by 100 to match the Fama-French data.
3. Convert both dates to datetime.
 - a. For Apple data specify the format as '%Y-%m-%d'.
 - b. For Fama-French data specify the format as '%Y%m%d'.
4. Merge the Apple returns and Fama-French data into a data frame named 'data'.
5. Drop all NAs.
6. Calculate the excess returns for Apple (return minus risk-free rate) in a new column.
7. Perform a linear regression using OLS with and without a constant (alpha). Interpret your results.

Exercise 2 – Linear Regression

In this exercise you will practice performing time series regression using OLS.

1. Load the SP500 volatility data (SP500_vol.csv).
2. Extract the 'rtnClose', 'vol', 'vol_1', 'vol_ma_5', 'vol_ma_21' columns into a data frame named ts500.
3. Scale the returns and volatility by 100 and 10000 respectively.
4. Add a constant to the data frame.
5. Perform a linear regression using OLS for the following equation:
$$Vol_t = \alpha + \beta_1 Vol_{t-1} + \beta_2 Vol_{MA5} + \beta_3 Vol_{MA21}$$
6. Interpret your results.
7. Graph the actual (y) and the estimated ($\hat{y}$) on the same chart.

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Advanced Questions

Exercise 3 – Portfolio Optimization

In this exercise you will practice performing portfolio optimization.

1. Load all the stock data except for the S&P500 data.
 - a. Resample to Monthly frequency.
 - b. Calculate simple returns for each stock using the adjusted close.
 - c. Combine the returns into a data frame where the index is date time and the columns are the individual stock returns.
2. Calculate the covariance matrix of the stocks.
3. Calculate the mean return of the stocks.
4. Scale the mean and covariance matrix by 100 and 10,000 respectively.
5. Find and plot the portfolio frontier for returns 1 to 10.

Exercise 4 – Principal Component Analysis

In this exercise you will practice performing a principal component analysis.

1. Load the Fama-French 25 portfolio data (ff25Z.csv).
2. Extract the return data into an array using the values property of the data frame.
3. Fit a PCA model to the return data. Use the name 'princomp'.
4. Examine the explained variance ratio, how much do the first 5 factors explain?
5. Extract the components from the PCA object into a variable named 'comp'.
 - a. Remember to take the transpose.
6. Rescale the weights for the 'comp' variable.
7. Find the first five factors.
8. Perform a Linear Regression for the 25 portfolios. What is the average R^2 of the regressions?